Interface agents and RO-Crates for a generic DBTL biomanufacturing workflow

Generic microbial biomanufacturing | Adapted from Carbonell et al., ***Communications Biology*** (2018)

Human researcher
Sets the target product, seeds the RO-Crates with constraints and approves rounds.

D → B | Design-to-Build Agent

- Checks designs and assembly feasibility against equipment limits
- Generates assembly recipes and robot worklists
- Reads all four RO-Crates and packages the Build RO-Crate

Handoff to Build: Build RO-Crate
- Part and plasmid designs (IDs)
- Assembly recipes and worklists
- Reduced library and build plan

Design (D)

Select pathway, enzymes and genetic parts for the target product.



Tools & techniques
- RetroPath and Selenzyme (pathway and enzyme selection)
- PartsGenie (RBS and codon optimisation)
- Design of experiments to reduce the library

Writes Design RO-Crate | Reads D B T L

Build (B)

Assemble and verify the strain or construct library.



Tools & techniques
- DNA synthesis and PCR part preparation
- Ligase cycling reaction assembly
- Worklist-driven robotic liquid handling
- Restriction digest and sequence verification

Writes Build RO-Crate | Reads D B T L

Shared RO-Crate store

All stages and agents read all four crates

Design RO-Crate
Seed: target product and host
Adds: parts, DoE library, rationale

Build RO-Crate
Seed: robots, vendors, plate formats
Adds: assembly log, QC, sequences

Coordinator
Tracks decisions, provenance, exceptions and human approvals; curates the RO-Crate store

Test RO-Crate
Seed: assay equipment and limits
Adds: raw MS data, QC flags, run log

Learn RO-Crate
Seed: model and compute limits
Adds: effects, models, design rules

Round 0: the researcher seeds each crate with constraints before the first cycle.

L → D | Learn-to-Design Agent

- Translates statistical effects and design rules into next-round constraints
- Reads all four RO-Crates and packages the Design RO-Crate

Handoff to Design: Design RO-Crate
- Design constraints (fixed factors)
- Prioritised factors and levels
- Next library specifications (e.g., full-factorial)

Learn (L)

Analyse results and derive design rules.



Tools & techniques
- Statistical analysis of design factors
- Machine learning models
- Design rules for the next round

Writes Learn RO-Crate | Reads D B T L

Test (T)

Cultivate strains and measure product titres.



Tools & techniques
- 96 deep-well growth and induction
- Automated product extraction
- UPLC-MS/MS quantification
- R scripts for data processing

Writes Test RO-Crate | Reads D B T L

B → T | Build-to-Test Agent

- Tracks sample IDs and barcodes from parts to strains
- Confirms assemblies by digest and sequencing, and flags failures
- Reads all four RO-Crates and packages the Test RO-Crate

Handoff to Test: Test RO-Crate
- Verified strains (IDs + sequences)
- Growth and extraction protocol
- Build QC status (pass/fail)

T → L | Test-to-Learn Agent

- Validates MS data, calibration curves and quality flags
- Prepares a curated titre dataset for modelling
- Reads all four RO-Crates and packages the Learn RO-Crate

Handoff to Learn: Learn RO-Crate
- Quantified titres (calibrated)
- QC report (controls, replicates)
- Metadata (strains, media, conditions)

Legend

- **DBTL stage** (primary experimental function)
- **Interface agent** (manages stage transitions)
- **Coordinator** (orchestration and decisions)
- **RO-Crate per stage** (one crate for each DBTL stage)
- **DBTL process flow** (Design → Build → Test → Learn)
- **Read access to all crates** (stages and agents)
- **Stage writes its own crate** (outputs and provenance)
- **Human input / approval** (seeding and round approval)

RO-Crate protocol

0 Seed
Before round 1, the researcher seeds the crates with constraints, e.g., available equipment.

1 Package
Each interface agent packages the next stage's crate with validated inputs.

2 Record
Each stage adds its outputs, IDs and provenance to its own crate.

3 Share
All stages and agents read all four crates, so Design can respect Build and Test limits.

Experimental tools from Carbonell et al. (2018) • Interface agents and RO-Crates are the proposed theoretical architecture, not used in the original pipeline.